Survey (Set 2)

When developers add new test cases into an existing test suite, do the developers look at the existing test cases and then reuse different attributes of existing test cases (such as variable naming, comments, test case naming, set of source code methods, overall test case organization/structure etc.) to derive the new testcases?

In order to answer the above question, we have designed this survey. The survey shows 20 pairs of unit test cases collected from open source projects. For each pair the participant is requested to indicate to what extent a developer *may/may not* have used a specific attribute to write the new test case.

Please answer all the following questions to complete the study. Thanks in advance.

* Indicates required question

Example 1: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
//-----------------------------------------------------------------------
public void testWrite_stringToOutputStream() throws Exception {
  String str = new String(inData, "US-ASCII");
  ByteArrayOutputStream baout = new ByteArrayOutputStream();
  YellOnFlushAndCloseOutputStream out = new YellOnFlushAndCloseOutputStream(baout, true, true);
  IOUtils.write(str, out);
  out.off();
  out.flush();
  assertEquals("Sizes differ", inData.length, baout.size());
  assertTrue("Content differs", Arrays.equals(inData, baout.toByteArray()));
}
//-----------------------------------------------------------------------
public void testWrite_charSequenceToOutputStream() throws Exception {
  CharSequence csq = new StringBuilder(new String(inData, "US-ASCII"));
  ByteArrayOutputStream baout = new ByteArrayOutputStream();
  YellOnFlushAndCloseOutputStream out = new YellOnFlushAndCloseOutputStream(baout, true, true);
  IOUtils.write(csq, out);
  out.off();
  out.flush();
  assertEquals("Sizes differ", inData.length, baout.size());
  assertTrue("Content differs", Arrays.equals(inData, baout.toByteArray()));
}
```

1. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

2. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

3. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

4. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

5. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

6. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

7. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

8. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

9. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 2: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
//    // TODO Remove after debugging
//    private void log(Object obj) {
//        System.out.println(
//            FileUtilsTestCase.class +"" "" + getName() + "" "" + obj);
//    }
// copyFile
public void testCopyFile1() throws Exception {
  File destination = new File(getTestDirectory(), ""
    copy1.txt "");
  //Thread.sleep(LAST_MODIFIED_DELAY);
  //This is to slow things down so we can catch if
  //the lastModified date is not ok
  FileUtils.copyFile(testFile1, destination);
  assertTrue(""
    Check Exist "", destination.exists());
  assertTrue(""
    Check Full copy "", destination.length() == testFile1Size);
  /* disabled: Thread.sleep doesn't work reliantly for this case
      assertTrue(""Check last modified date preserved"",
          testFile1.lastModified() == destination.lastModified());*/
}

public void testMoveFile_Rename() throws Exception {
  File destination = new File(getTestDirectory(), ""
    move1.txt "");
  FileUtils.moveFile(testFile1, destination);
  assertTrue(""
    Check Exist "", destination.exists());
  assertTrue(""
    Original deleted "", !testFile1.exists());
}
```

10. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

11. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

12. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

13. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

14. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

15. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

16. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

17. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

18. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 3: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void testGroupOrAttribute() {
    String h = ""<div id=1 /><div id=2 /><div title=foo /><div title=bar />"";
    Elements els = Jsoup.parse(h).select(""[id],[title=foo]"");
    assertEquals(3, els.size());
    assertEquals(""1"", els.get(0).id());
    assertEquals(""2"", els.get(1).id());
    assertEquals(""foo"", els.get(2).attr(""title""));
}
@Test
public void deeperDescendant() {
    String h = ""<div class=head><p class=first>Hello</div>
                <div class=head><h1 class=first>Another</h1><p>Again</div>"";
    Elements els = Jsoup.parse(h).select(""div p .first"");
    assertEquals(1, els.size());
    assertEquals(""Hello"", els.get(0).text());
}
```

19. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

20. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

21. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

22. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

23. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

24. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

25. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

26. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

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◯	◯	◯	◯	◯

27. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 4: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
/**
 * Test Filtering
 */
public void testFilter() {
    List<File> results = new TestFileFinder(dirsAndFilesFilter, -1).find(javaDir);
    assertEquals(""Result Size"", (1 + dirs.length + ioFiles.length + outputFiles.length), results.size());
    assertTrue(""Start Dir"", results.contains(javaDir));
    checkContainsFiles(""Dir"", dirs, results);
    checkContainsFiles(""IO File"", ioFiles, results);
    checkContainsFiles(""Output File"", outputFiles, results);
}
/**
 * Test Filtering
 */
public void testFilterString() {
    List<String> results = new TestFileFinderString(dirsAndFilesFilter, -1).find(javaDir);
    assertEquals(""Result Size"", (outputFiles.length + ioFiles.length), results.size());
    checkContainsString(""IO File"", ioFiles, results);
    checkContainsString(""Output File"", outputFiles, results);
}"
```

28. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

29. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

30. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

31. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

32. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

33. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

34. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

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◯	◯	◯	◯	◯

35. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

36. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 5: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
/**
 * Test the comparator array sort.
 */
public void testSortArray() {
    File[] files = new File[3];
    files[0] = equalFile1;
    files[1] = moreFile;
    files[2] = lessFile;
    comparator.sort(files);
    assertTrue(""equal"", files[0] == lessFile);
    assertTrue(""less"", files[1] == equalFile1);
    assertTrue(""more"", files[2] == moreFile);
}
/**
 * Test comparator list sort is null safe.
 */
public void testSortListNull() {
    assertNull(comparator.sort((List) null));
}
```

37. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

38. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

39. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

40. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

41. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

42. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

43. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

44. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

45. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 6: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
"@Test
public void testBasicEncodeDecode() throws Exception {
    final QuotedPrintableCodec qpcodec = new QuotedPrintableCodec();
    final String plain = "= Hello there =\r\n";
    final String encoded = qpcodec.encode(plain);
    assertEquals("Basic quoted-printable encoding test", "=3D Hello there
        =3D=0D=0A", encoded);
    assertEquals("Basic quoted-printable decoding test", plain, qpcodec
        .decode(encoded));
}
@Test
public void testTrailingSpecial() throws Exception {
    final QuotedPrintableCodec qpcodec = new QuotedPrintableCodec(true);
    String plain = "This is a example of a quoted-printable text file. This
        might contain sp=cial chars.";
    String expected = "This is a example of a quoted-printable text file.
        This might contain sp=3D=\r\ncial chars.";
    assertEquals(expected, qpcodec.encode(plain));
    plain = "This is a example of a quoted-printable text file. This might
        contain ta\tbs as well.";
    expected = "This is a example of a quoted-printable text file. This
        might contain ta=09=\r\nbs as well.";
    assertEquals(expected, qpcodec.encode(plain));
}"
```

46. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

47. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

48. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

49. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

50. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

51. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

52. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

53. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

54. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 7: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
// copyURLToFile
public void testCopyURLToFile() throws Exception {
    // Creates file
    File file = new File(getTestDirectory(), getName());
    file.deleteOnExit();
    // Loads resource
    String resourceName = ""/java/lang/Object.class"";
    FileUtils.copyURLToFile(getClass().getResource(resourceName), file);
    // Tests that resuorce was copied correctly
    FileInputStream fis = new FileInputStream(file);
    try {
        assertTrue(""Content is not equal."", IOUtils.contentEquals(getClass().getResourceAsStream
            (resourceName), fis));
    } finally {
        fis.close();
    }
    //TODO Maybe test copy to itself like for copyFile()
}
public void testCopyURLToFileWithTimeout() throws Exception {
    // Creates file
    File file = new File(getTestDirectory(), ""testCopyURLToFileWithTimeout"");
    file.deleteOnExit();
    // Loads resource
    String resourceName = ""/java/lang/Object.class"";
    FileUtils.copyURLToFile(getClass().getResource(resourceName), file, 500, 500);
    // Tests that resuorce was copied correctly
    FileInputStream fis = new FileInputStream(file);
    try {
        assertTrue(""Content is not equal."", IOUtils.contentEquals(getClass().getResourceAsStream
            (resourceName), fis));
    } finally {
        fis.close();
    }
    //TODO Maybe test copy to itself like for copyFile()
}
```

55. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

56. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

57. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

58. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

59. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

60. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

61. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

62. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

63. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 8: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
/**
 * Test writing Null String
 */
public void testNullString() {
    ProxyWriter proxy = new ProxyWriter(new NullWriter());
    try {
        proxy.write((String) null);
    } catch (Exception e) {
        fail(""Writing null String threw "" + e);
    }
    try {
        proxy.write((String) null, 0, 0);
    } catch (Exception e) {
        fail(""Writing null String threw "" + e);
    }
}
/**
 * Test Writing a Partial String
 */
public void testWriteStringPartial() {
    StringBuilderWriter writer = new StringBuilderWriter();
    ProxyWriter proxy = new ProxyWriter(writer);
    try {
        proxy.write(""ABC"", 1, 2);
    } catch (Exception e) {
        fail(""Writing String threw "" + e);
    }
    assertEquals(""BC"", writer.toString());
}
```

64. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

65. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

66. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

67. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

68. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

69. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

70. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

71. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

72. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 9: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void testDecodeStringOddCharacters() {
    try {
        new Hex().decode("6");
        fail("An exception wasn't thrown when trying to decode an odd number of
            characters");
    } catch (final DecoderException e) {
        // Expected exception
    }
}
@Test
public void testDecodeStringEmpty() throws DecoderException {
    assertTrue(Arrays.equals(new byte[0], (byte[]) new Hex().decode("")));
}
```

73. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

74. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

75. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

76. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

77. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

78. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

79. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

80. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

81. Please indicate if any other attributes were used (not listed in above)

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Untitled Section

Example 10: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
"public void testLargeUTF8WithBufferedWrite() throws IOException {
    testWithBufferedWrite(LARGE_TEST_STRING, ""UTF-8"");
}
public void testLargeUTF16WithBufferedWrite() throws IOException {
    testWithBufferedWrite(LARGE_TEST_STRING, ""UTF-16"");
}"
```

82. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

83. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

84. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

85. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

86. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

87. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

88. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

89. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

90. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 11: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
@Test
public void testBasicEmpty2() {
    final String input = "a  b c";
    final StrTokenizer tok = new StrTokenizer(input);
    tok.setIgnoreEmptyTokens(false);
    tok.setEmptyTokenAsNull(true);
    assertEquals("a", tok.next());
    assertEquals(null, tok.next());
    assertEquals("b", tok.next());
    assertEquals("c", tok.next());
    assertFalse(tok.hasNext());
}
@Test
public void testDelimMatcherQuoteMatcher() {
    final String input = "`a`;`b`;`c`";
    final StrMatcher delimMatcher = new StrMatcher.CharSetMatcher(new char[] { ';' });
    final StrMatcher quoteMatcher = new StrMatcher.CharSetMatcher(new char[] { '`' });
    final StrTokenizer tok = new StrTokenizer(input, delimMatcher, quoteMatcher);
    assertEquals("a", tok.next());
    assertEquals("b", tok.next());
    assertEquals("c", tok.next());
    assertFalse(tok.hasNext());
}
```

91. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

92. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

93. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

94. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

95. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

96. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

97. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

98. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

99. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 12: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
/**
 * Tests constructor.
 */
@Test
public void testConstructorMapPrefixSuffix() {
    final Map<String, String> map = new HashMap<>();
    map.put(""name"", ""commons"");
    final StrSubstitutor sub = new StrSubstitutor(map, ""<"", ""&gt;"")
    assertEquals(""Hi < commons"", sub.replace(""Hi $< <name>""));
}
@Test
public void testReplaceInTakingStringBuilderWithNull() {
    Map<String, Object> map = new HashMap<>();
    StrSubstitutor strSubstitutor =
            new StrSubstitutor(map, """", """", 'T', ""K+<'f"");
    assertFalse(strSubstitutor.replaceIn((StringBuilder) null));
}
```

100. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

101. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

102. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

103. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

104. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

105. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

106. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

107. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

108. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 13: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
public void testWriteStringToFile2() throws Exception {
    File file = new File(getTestDirectory(), "write.txt");
    FileUtils.writeStringToFile(file, "Hello /u1234", (String) null);
    byte[] text = "Hello /u1234".getBytes();
    assertEqualContent(text, file);
}
public void testWriteStringToFile3() throws Exception {
    File file = new File(getTestDirectory(), "write.txt");
    FileUtils.writeStringToFile(file, "Hello /u1234", (Charset) null);
    byte[] text = "Hello /u1234".getBytes();
    assertEqualContent(text, file);
}
```

109. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

110. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

111. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

112. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

113. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

114. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

115. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

116. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

117. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 14: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void descendant() {
    String h = ""<div class=head><p class=first>Hello</p><p>There</p></div><p>None</p>
    Document doc = Jsoup.parse(h);
    Elements els = doc.select("".head p"");
    assertEquals(2, els.size());
    assertEquals(""Hello"", els.get(0).text());
    assertEquals(""There"", els.get(1).text());
    Elements p = doc.select(""p.first"");
    assertEquals(1, p.size());
    assertEquals(""Hello"", p.get(0).text());
}
@Test
public void parentChildElement() {
    String h = ""<div id=1><div id=2><div id = 3></div></div></div><div id=4></div>"";
    Document doc = Jsoup.parse(h);
    Elements divs = doc.select(""div > div"");
    assertEquals(2, divs.size());
    // 2 is child of 1
    assertEquals(""2"", divs.get(0).id());
    // 3 is child of 2
    assertEquals(""3"", divs.get(1).id());
    Elements div2 = doc.select(""div#1 > div"");
    assertEquals(1, div2.size());
    assertEquals(""2"", div2.get(0).id());
}
```

118. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

119. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

120. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

121. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

122. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

123. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

124. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

125. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

126. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 15: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
"//-----------------------------------------------------
@Test
public void testDeleteCharAt() {
    final StrBuilder sb = new StrBuilder(""abc"");
    sb.deleteCharAt(0);
    assertEquals(""bc"", sb.toString());
    try {
        sb.deleteCharAt(1000);
        fail(""Expected IndexOutOfBoundsException"");
    } catch (final IndexOutOfBoundsException e) {
    }
}
@Test(expected = StringIndexOutOfBoundsException.class)
public void testDeleteCharAtWithNegative() {
    new StrBuilder().deleteCharAt((-1258));
}"
```

127. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

128. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

129. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

130. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

131. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

132. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

133. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

134. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

135. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 16: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
"// ----------------------------------------------------------------------
@Test
public void testAbbreviateForLowerValueAndAppendedString() {
    assertEquals(""012"", WordUtils.abbreviate(""012 3456789"", 0, 5, null));
    assertEquals(""01234-"", WordUtils.abbreviate(""01234 56789"", 5, 10, ""-""));
    assertEquals(""01 23 45 67abc"", WordUtils.abbreviate(""01 23 45 67 89"", 9, -1, ""abc""));
    assertEquals(""01 23 45 6"", WordUtils.abbreviate(""01 23 45 67 89"", 9, 10, ""));
}
@Test(expected = IllegalArgumentException.class)
public void testAbbreviateForLowerThanMinusOneValues() {
    assertEquals(""01 23 45 67"", WordUtils.abbreviate(""01 23 45 67 89"", 9, -10, null));
}"
```

136. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

137. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

138. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

139. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

140. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

141. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

142. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

143. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

144. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 17: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void testExcelPrintAllArrayOfArrays() throws IOException {
    final StringWriter sw = new StringWriter();
    final CSVPrinter printer = new CSVPrinter(sw, CSVFormat.EXCEL);
    printer.printRecords(new String[][] { { "r1c1", "r1c2" }, { "r2c1", "r2c2" } });
    assertEquals("r1c1,r1c2" + lineSeparator + "r2c1,r2c2" + lineSeparator, sw.toString());
}
@Test
public void testPrinterQuoteAll() throws IOException {
    final StringWriter sw = new StringWriter();
    final CSVPrinter printer = new CSVPrinter(sw, CSVFormat.DEFAULT.withQuotePolicy(Quote.ALL));
    printer.printRecord("a", "b\nc", "d");
    assertEquals("\"a\",\"b\nc\",\"d\"" + lineSeparator, sw.toString());
```

145. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

146. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

147. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

148. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

149. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

150. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

151. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

152. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

153. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 18: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
"@Test
public void descendant() {
    String h = "<div class=head><p class=first>Hello</p><p>There</p
        ></div><p>None</p>";
    Document doc = Jsoup.parse(h);
    Elements els = doc.select(".head p");
    assertEquals(2, els.size());
    assertEquals("Hello", els.get(0).text());
    assertEquals("There", els.get(1).text());
    Elements p = doc.select("p.first");
    assertEquals(1, p.size());
    assertEquals("Hello", p.get(0).text());
}
@Test
public void deeperDescendant() {
    String h = "<div class=head><p class=first>Hello</div><div class
        =head><h1 class=first>Another</h1><p>Again</div>";
    Elements els = Jsoup.parse(h).select("div p .first");
    assertEquals(1, els.size());
    assertEquals("Hello", els.get(0).text());
}
```

154. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

155. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

156. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

157. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

158. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

159. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

160. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

161. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

162. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 19: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Override
@Test
public void testGetFile() {
    // Cannot convert a Writer to a File.
    assertThrows(UnsupportedOperationException.class, super::testGetFile);
}
@Override
@Test
public void testSize() {
    // Cannot convert a Writer to a size.
    assertThrows(UnsupportedOperationException.class, super::testSize);
}
```

163. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

164. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

165. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

166. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

167. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

168. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

169. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

170. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

171. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 20: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void testGet() {
    assertThrows(UncheckedIOException.class, () -> Uncheck.get(() -> {
        throw new IOException();
    }));
    assertThrows(UncheckedIOException.class, () -> Uncheck.get(TestConstants
        .THROWING_IO_SUPPLIER));
    assertEquals("new1", Uncheck.get(() -> TestUtils.compareAndSetThrowsIO(ref1, "new1")));
    assertEquals("new1", ref1.get());
}
@Test
public void testGetAsInt() {
    assertThrows(UncheckedIOException.class, () -> Uncheck.getAsInt(() -> {
        throw new IOException();
    }));
    assertThrows(UncheckedIOException.class, () -> Uncheck.getAsInt(TestConstants
        .THROWING_IO_INT_SUPPLIER));
    assertEquals(1, Uncheck.getAsInt(() -> TestUtils.compareAndSetThrowsIO(atomicInt, 1)));
    assertEquals(1, atomicInt.get());
}
```

172. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

173. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

174. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

175. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

176. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

177. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

178. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

179. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

180. Please indicate if any other attributes were used (not listed in above)

_____

Demographic Information

181. What is you current post * Dropdown

Mark only one oval.

- PhD Student
- Masters Student
- Undergraduate Student
- Software Developer
- Others

182. How many years of programming experience you have? Dropdown

Mark only one oval.

- 0-1 year
- 1-3 years
- 3-5 years
- more than 5 years

183. Do you have experience in writing unit test cases?

Mark only one oval.

- Yes
- No
- Maybe

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